Semantic Governance

Walkaround Deck



What is Semantic Governance?

SEMANTIC GOVERNANCE

WHAT:

Semantic Governance helps users access data using **familiar terms** (not necessarily those in the glossary of terms in a catalog or perhaps an enterprise dictionary). Semantics are associated with a point of view (**POV**) for a persona and/or a point in context (**scenario**). But, it's more than just simple mapping of terms. It extends to phrases that can be interpreted and mapped to complex data. The semantic layer provides contextual relevance for our analytics so we can harmonize points of view into a holistic and consistent understanding.

HOW:

In order to achieve a consistent meaning of content it must be informed by context that is relevant (mapped to terms that are familiar to Personas) across the organization. Requires a disciplined effort to curate both the semantics and their mappings to our structured vocabularies and ontologies.

Business Focus

Why invest in a Semantic Governance?

SEMANTIC GOVERNANCE

What can it do for me?

Aggregates, integrates, and organizes data and information to create a unified view, making it easily accessible for decision-makers by providing relevant insights and context at the right time.

Provides a structured way to represent information to model relationships, context, and hierarchies to enhance learning, curate context, and connect structured and unstructured data to help AI understand and interpret data and information to deliver meaningful insights.

Supports continuous learning for AI so it can adapt as new data, context, and information become available and improves insight delivery by enhancing transparency and explainability.

Business Focus

How will Semantic Governance help?

SEMANTIC GOVERNANCE

Key Outcomes:

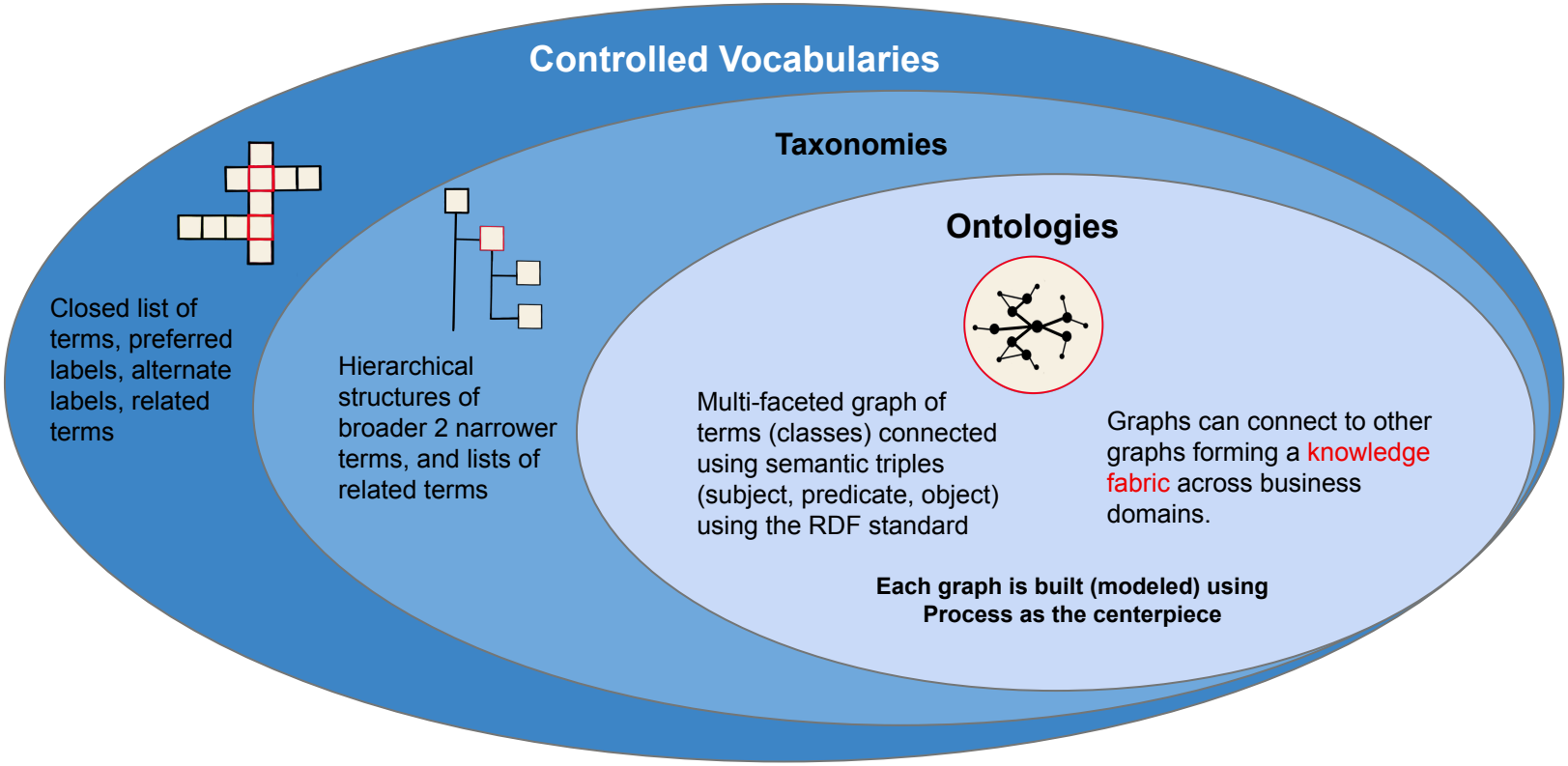
Consistent | Connected | Clear understanding about what changed, why things changed, and who might be impacted. Ensure our Data is FAIR (next slide) and Consumable (search & understand).

We could leverage the semantic layer to **reduce ambiguity** with different reports about the same topic (for example: Customer Churn could generate multiple reports showing various perspectives, where each is correct but only result in confusion until they are contextually harmonized so we can clarify meaning across a complex set of factors.)

Goal:

Deliver **adaptable** services, processes, systems, analytics, UX, and CX and **accelerate decision making** across the organization.

Semantic Governance Scope



Business Focus

Semantic Governance: Target Outcomes

SEMANTIC GOVERNANCE

Trusted BI & AI
Smarter, More Reliable AI & Analytics

Business Literacy
A Single Source of Truth for Business Language

Meaning at Scale
Future-Proof, Standards-Based Knowledge

Improved Decision Making
More Informed Decisions, Faster

Knowledge Preservation
Living Business Knowledge That Evolves With Us

Align Business with Technology
One Semantic Layer Connecting People and Systems

Business Focus

Aim Higher: Enabling Progressive Search

SEMANTIC GOVERNANCE

Basic Search

Keyword matching, boolean logic, simple filtering, and personalized results based on Persona Outcomes.

1

We are here

2

Augmented Search

Support for complex logic and phrase matching, basic analytics, and personalized analytics views based on user roles and preferences.

Intelligent Search

Multi-dimensional analytics, causal analysis, trend forecasting, and AI-powered insights based on individual goals and user context.

3

4

Simulation Search

Simulate outcomes based on user-defined scenarios, model "what-if" scenarios, and highlight relevant simulations based on past actions.

Intelligent Assistant

Natural language processing for conversational search, real-time decision support, and personalized recommendations based on context, history, and intent.

5

Where we need to be

Business Focus

Governance Dashboard: Areas, Standards, Engineering

Governance Practice	Area Focused	Standards (ISO/OMG)	Supporting Engineering Discipline
DG Data Governance	Data quality, access	ISO 38505, DAMA DMBOK, ISO 8000	Data Engineering, Information Systems
Part of DG Master Data Management	Core entities, harmonization	OMG MDM, ISO 11179	Database Engineering, Systems Design
SG Semantic Governance	Vocabularies, Taxonomies, Ontologies, Meaning, Context, Lineage	W3C, OMG Semantic Governance, ISO 11179	Knowledge Engineering
Part of SG Metadata Management	Data Context, Lineage	ISO 11179, OMG Meta-Object Facility	Software Engineering, Information Architecture
BI Governance	Reporting, analytics	Various (industry-specific)	Business Intelligence, Analytics Engineering
AI Governance	Model oversight, ethics	ISO/IEC 42001, EU AI Act, NIST AI RMF	Machine Learning Engineering, Responsible AI
DI Decision Intelligence Governance	Decision modeling, feedback	OMG Decision Model & Notation (DMN)	Decision Science, Operations Research

Governance Focus

Governance Dashboard: What, Why, HOW SG Impacts

Governance Type	WHAT is Governed	WHY it Matters	HOW Semantic Governance Impacts
DG Data Governance	Data quality, access, privacy, lifecycle, stewardship	Ensures trustworthy, compliant, and usable data	Provides unified meaning, links policies to semantics, enables interoperability
Part of DG Master Data Management	Core business entities ("golden records")—customers, products, etc.	Accurate foundational data powers business operations, analytics, compliance	Aligns entity definitions, links master data across silos via ontologies, enables consistency
SG Semantic Governance	Meanings, ontologies, taxonomies, cross-organizational alignment	Prevents data silos, boosts interoperability, ensures context and meaning	Directly governs meanings, creates harmonized semantic artefacts, influences all other governance areas
Part of SG Metadata Management	Data about data—context, lineage, definitions	Traceability, discoverability, regulatory reporting, AI & analytics success	Aligns metadata vocabularies, integrates glossaries, harmonizes context for advanced analytics
BI Governance	Analytics, business reporting, KPIs, dashboard data	Accurate decision-making, regulatory reporting, performance management	Links report definitions to semantic models and controlled vocabularies for consistency
AI Governance	Models, training data, model ethics and transparency	Mitigates risk, ensures responsible use, explains outcomes and compliance	Embeds semantic intelligence in models, standardizes input/output concepts, improves explainability
DI Decision Intelligence Governance	Decision making frameworks, feedback loops, rationale, decision models	Increases decision quality, traceability, accountability, optimizes outcomes	Links decision logic, business rules, and causal relationships through semantic artifacts and ontology alignment

Governance Focus

Enabling the Knowledge Fabric

SEMANTIC GOVERNANCE

Data Fabric

- Focus on infrastructure enabling integration, access, and orchestration of data across diverse systems, formats, and locations in a unified, scalable way.
- Ensure that data is accessible, governed, and available to support operational and analytical use cases.
- Typically includes ETL/ELT tools, data lakes, data warehouses, APIs, and governance frameworks

Complimentary

Knowledge Fabric

- Focus on context, meaning, and intelligence from integrated data.
- Extends the data fabric emphasizing the semantic layer by dynamically enriching data with context, relationships, and meaning using ontologies, knowledge graphs, and AI-driven reasoning.
- Derives insights, enhances decision-making, and creates a shared understanding across an organization by connecting and contextualizing data.

Governance Focus

Semantic Governance Overview

Semantic Governance is built upon standards (ISO) and best practices for Knowledge Engineering to create and maintain the core knowledge (vocabularies, taxonomies, and ontologies) shared across the organization and the domain-specific knowledge (vocabularies, taxonomies, and ontologies unique to each domain and sub-domain)

These knowledge artifacts are built and maintained in our Semantic Knowledge Layer using the Semantic Services (prior slide) and governed with our Operating Model to ensure controls are in place covering all phases of the Knowledge Engineering Lifecycle including publishing and downstream consumption of enterprise and domain-specific knowledge.

Governance Focus

Semantic Governance (What & Where)

WHAT is it?

- Semantic Governance is built atop ontologies, linked data, and graphs to simplify search, discovery, and enable advanced querying, reasoning, and understanding across digital assets.

WHERE does it fit in?

- Semantic Governance is enabled by a Semantic Knowledge Layer that sits atop existing infrastructure (e.g., data, systems, tools) to provide a unified, meaningful view of structured and unstructured information for services and stakeholders.

Goal: Improve Relevancy & Understanding for Business and Technical Stakeholders

Governance Focus

Semantic Governance (How)

Establish Foundation, Operating Model and Semantic Services to **reduce ambiguity, manage heterogeneity, and improve the accuracy and relevancy of search for people and machines** (including AI via Vector DB & LLM optimization).

Semantic Services covers 3 core areas:

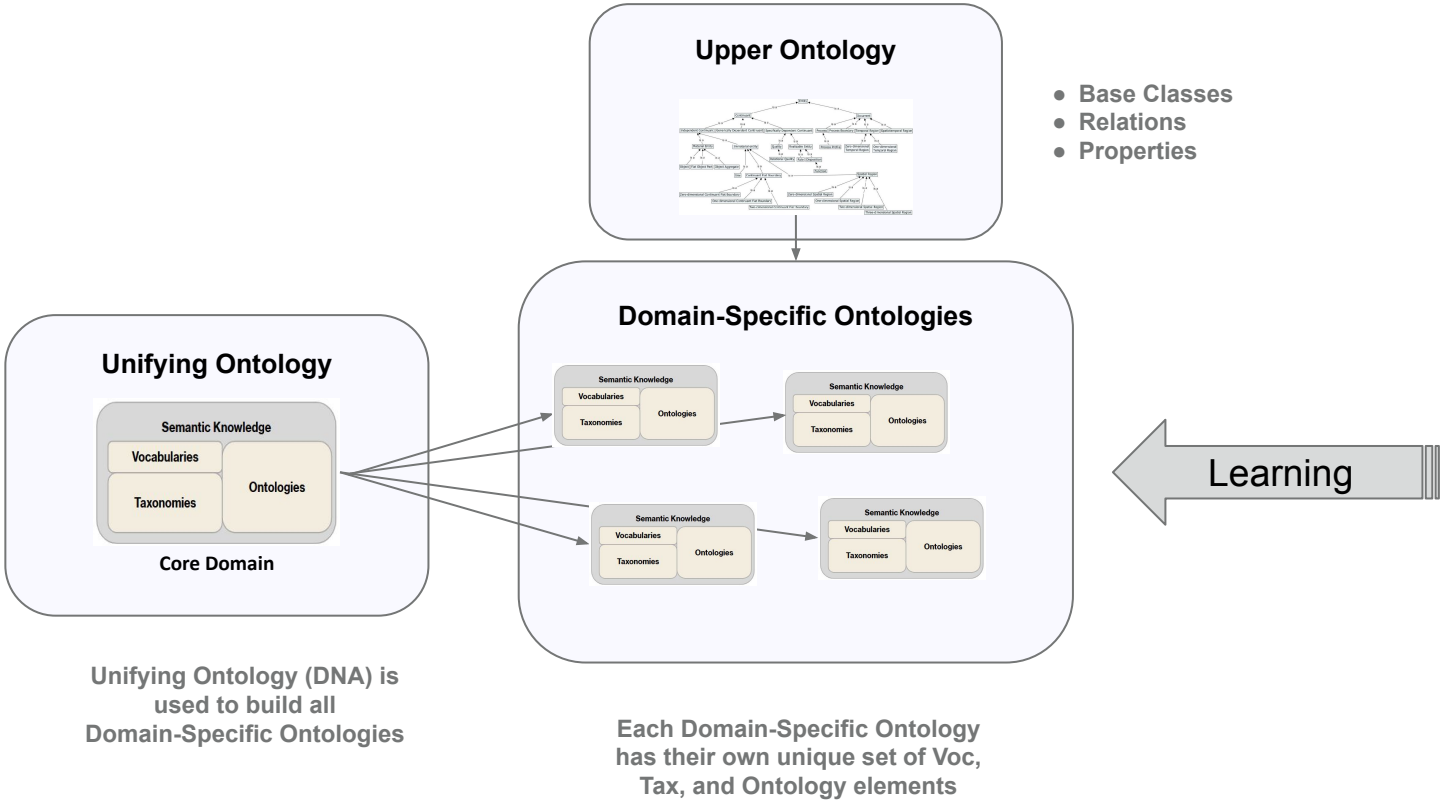
- Inbound - ingestion of structures which contain metadata about Digital Assets across the company
- Processing - mapping of knowledge (Top-Down) to structures for digital assets (Bottom-Up) and extending knowledge (Top-Down)
- Outbound - publishing Vocabularies, Taxonomies, Ontologies as Knowledge Artifacts that can be used by systems to deliver value

Governance Focus

Technical Deep Dive

Implementing Semantic Governance Services

Semantic Governance (Top-Down Structure)



Semantic Governance (Bottom Up Structure)

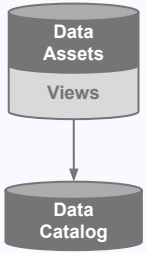
SEMANTIC GOVERNANCE

Compass



Learning

Governed Content



Ingestion Pipeline
(Semantic Services)

Metaphactory



Schema Ontologies

Ontology
Ontology
Ontology
Ontology
Ontology
Ontology

Class Objects representing
each structure (DB, Table, Cols)

- Definitions
- Attributes
- Schema (versions)



- Rag-Based Learning
- AI Training Dashboard

Semantic Governance (Mapping Structure)

SEMANTIC GOVERNANCE

Compass

Approved LLMs

Claude — Safety-focused and efficient text-based model.

CoPilot — Enterprise-ready AI assistance integrated across Microsoft applications, known for deep productivity and chat capabilities.

Domain-Specific Ontologies

Semantic Knowledge: Vocabularies, Taxonomies, Ontologies

Schema Ontologies

Ontology

Metaphactory



Semantic Search

Schema Mapping Ontologies

Semantic Knowledge: Vocabularies, Taxonomies, Ontologies

Ontology

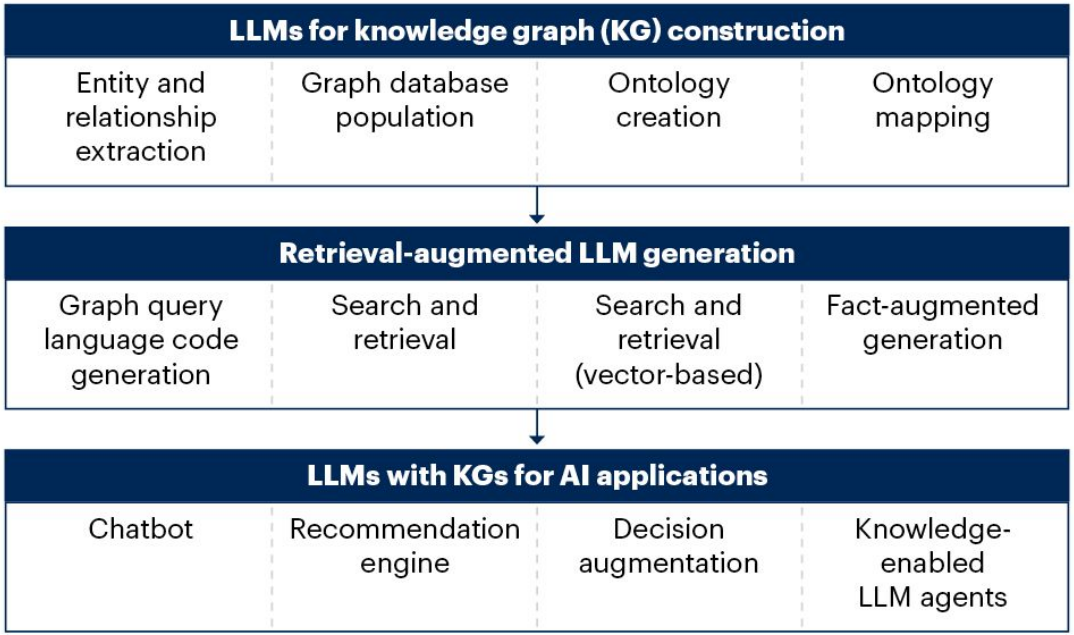
Top-Down Ontologies are mapped to Bottom Up Ontologies



- Rag-Based Learning
- AI Training Dashboard

Extending Semantic Governance with AI: Leveraging LLMs

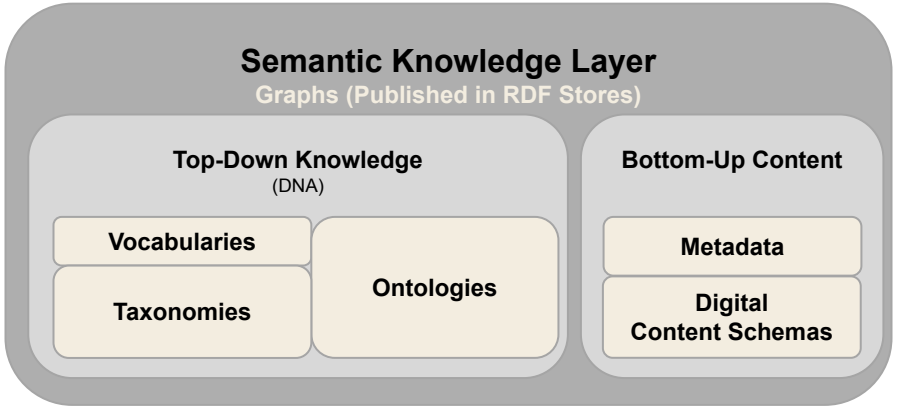
Knowledge-Graph-Augmented Large Language Model (LLM) Design Patterns



Source: Gartner
830998_C

Gartner

Conceptual View: Mapping Meaning to Digital Assets



Now

The 3-Tier Approach to Smarter Content & Knowledge

Uses current infrastructure to make content more relevant to users based on their context and persona.

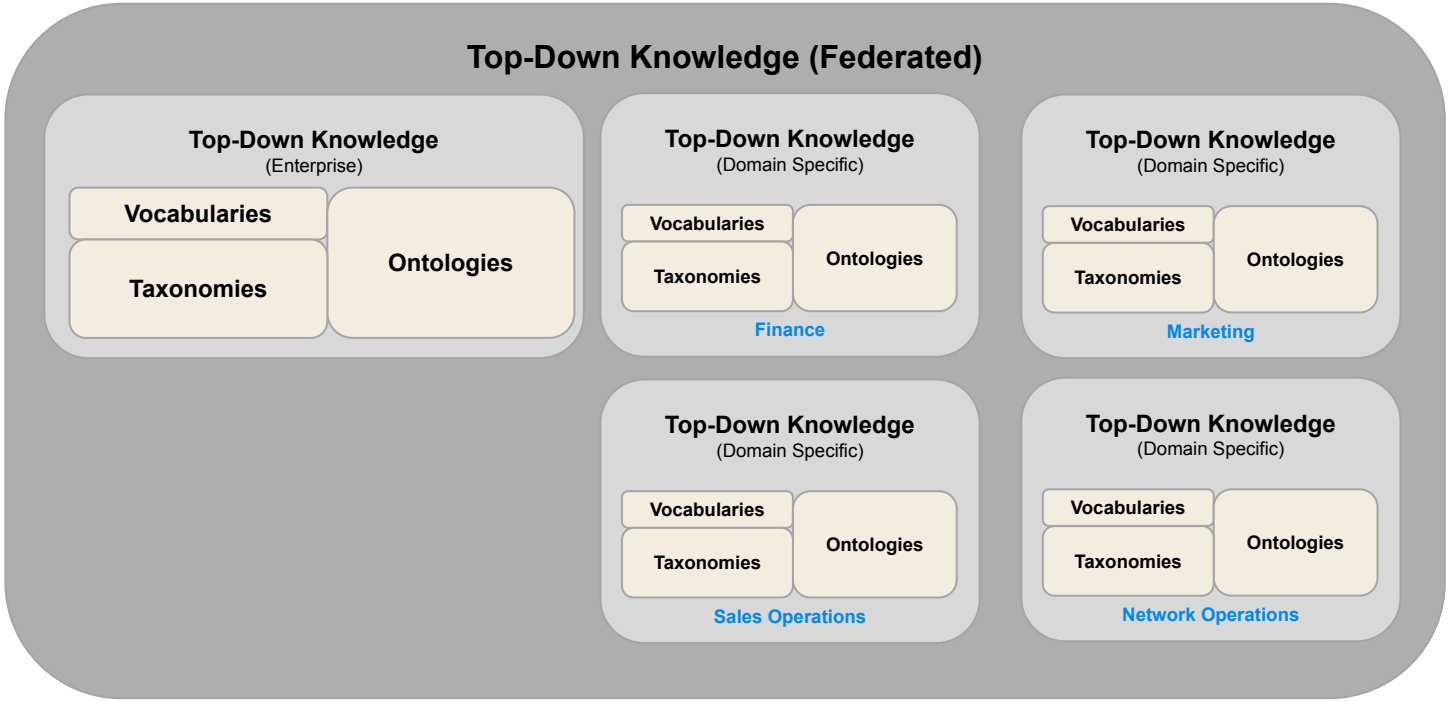
- Smarter organization by breaking down complex topics into manageable component knowledge
- Unified intelligence by establishing clear rules for a more evolved model of knowledge
- Enhance search capabilities

Future

Building a Learning System

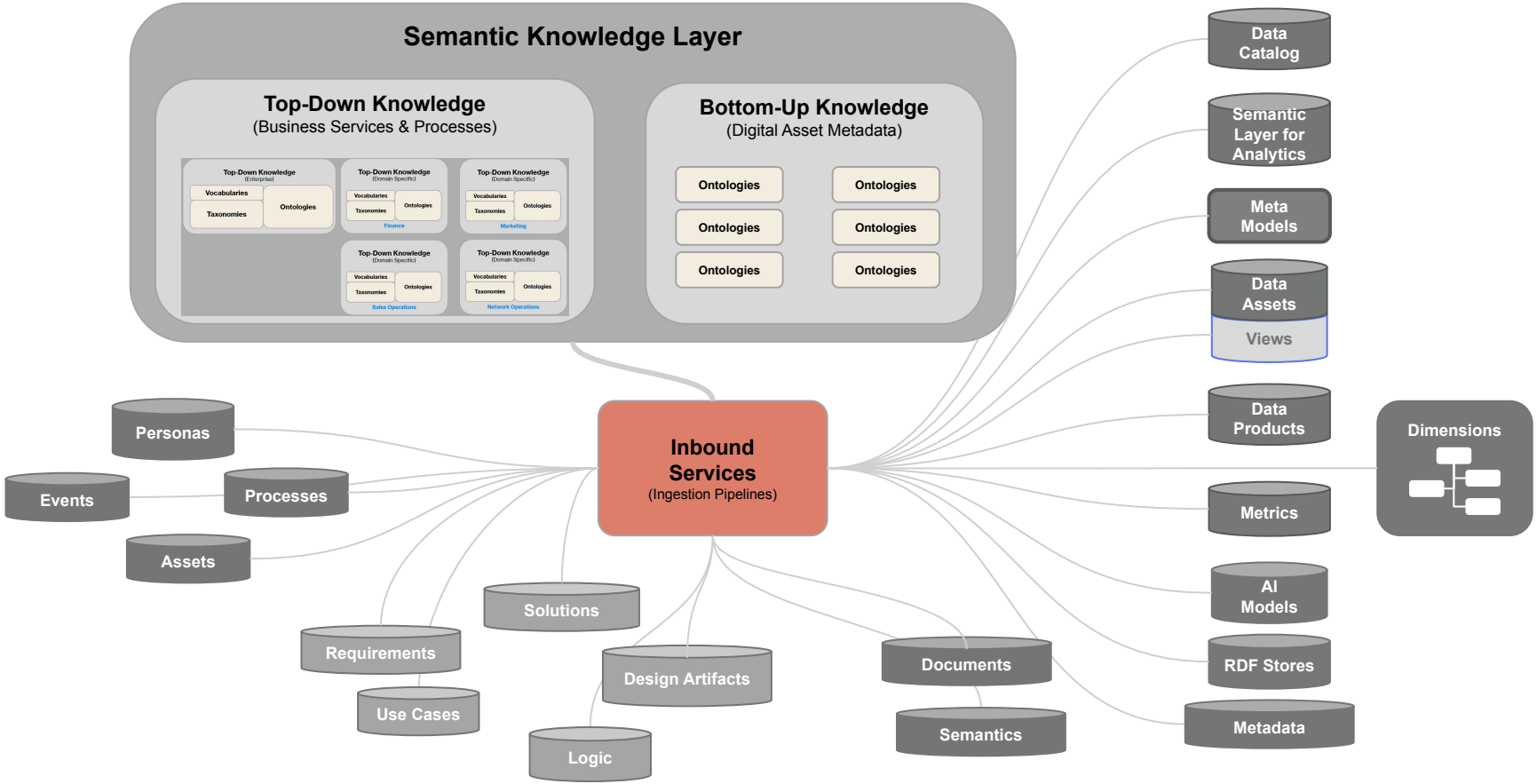
- Introduce lifecycle standards and design patterns
- Implement both virtual & physical knowledge graphs
- Develop an evolving learning capability by capturing clear facts & capturing probable knowledge

Component Model for Knowledge: Overview

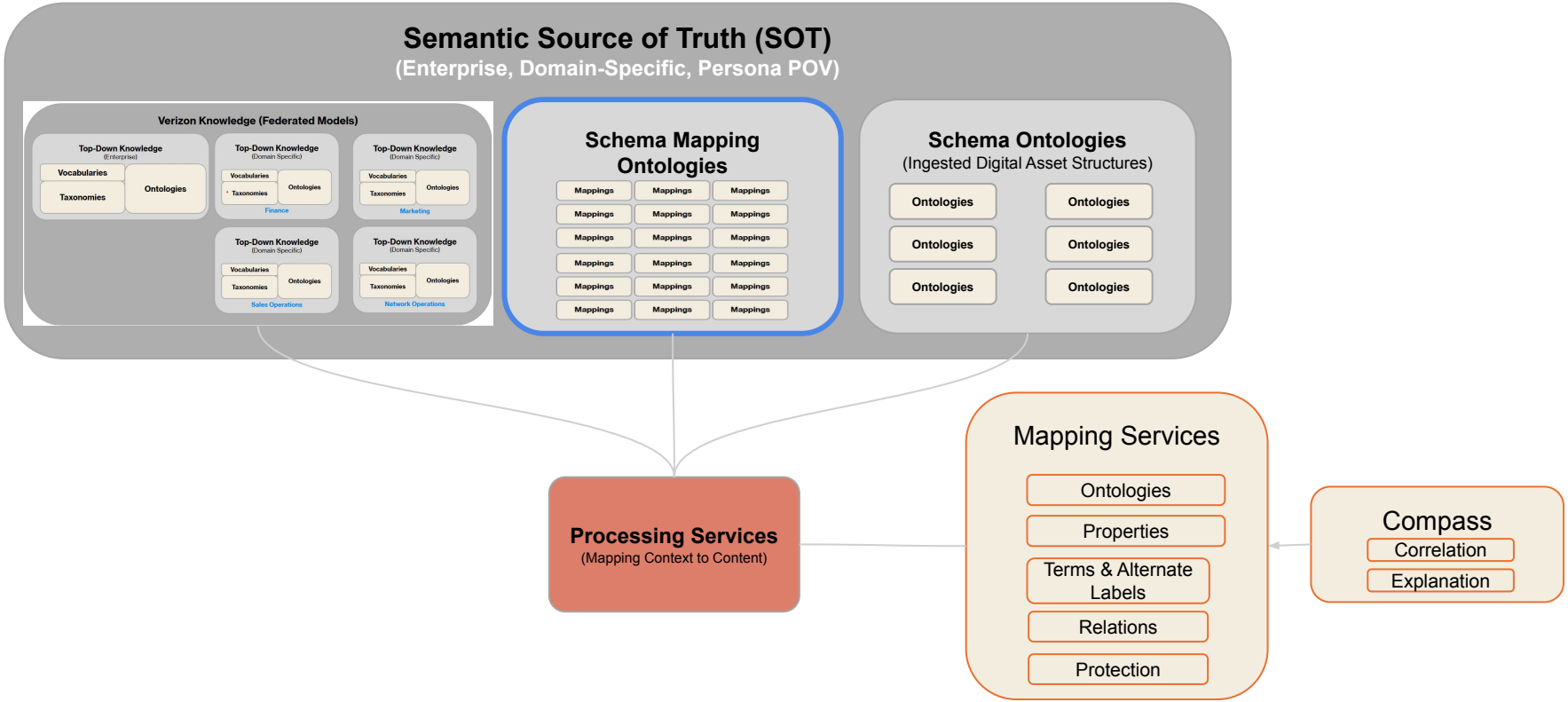


Component Design for Building out Company Knowledge

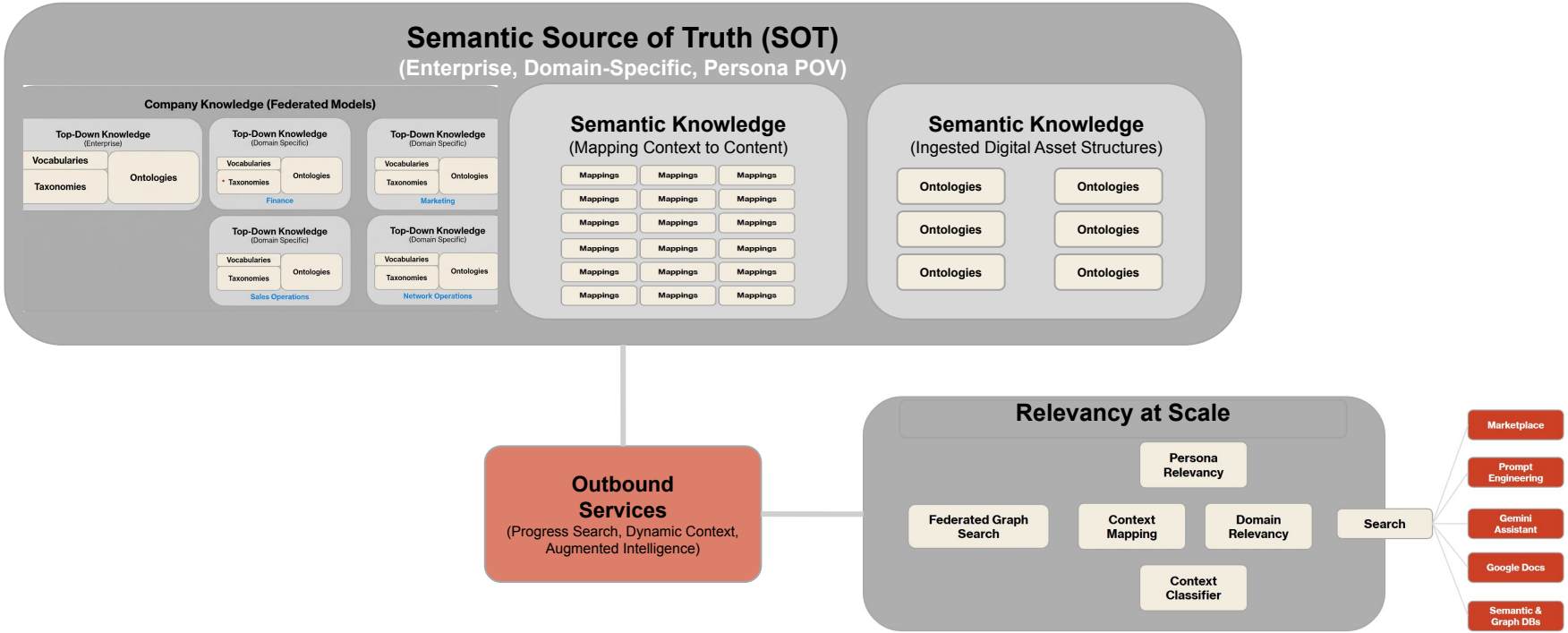
Semantic Services: Ingest Metadata about Digital Assets



Semantic Services: Map Context to Content (Meaning at Scale)



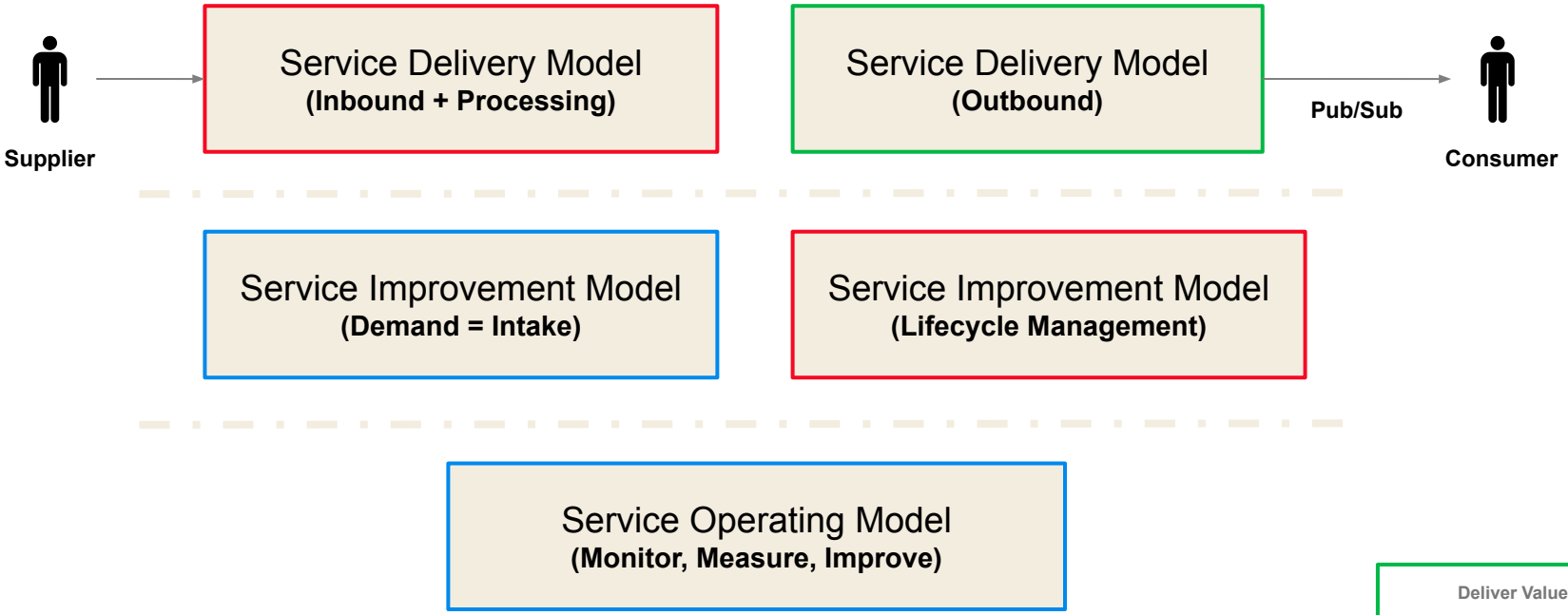
Semantic Services: Push Knowledge and/or Pull Relevant Search



Appendix

Supporting Materials

Semantic Governance (Service Management)



Semantic Governance (Service Management)

Service Delivery Model

INBOUND

- Sourcing
- Ingestion
- Scheduling

PROCESSING

- Mapping at Scale
  - SKL to Digital Assets (Vertically)
  - SKL to 3rd Party (Horizontally)
- Training & Learning
- Extending Knowledge
  - Manually
  - Automated

OUTBOUND

- Publishing
- Subscription

Service Improvement Model

DEMAND

- Customer Requests
- Service Gaps
- Capability Roadmap

DEFINE → **DESIGN** → **DEVELOP** → **DELIVER**

- Knowledge Engineering
- Systems Engineering
- Information Architecture
- W3C, OMG Semantic Governance
- ISO 11179, OMG Meta-Object Facility

Service Operating Model

- Service Level Management
- Monitoring & Measurement
- Continuous Improvement
- Continuous Learning
- Business Continuity
- Issue / Investigation Management
- Escalation/ Communications Management
- Problem / Resolution Management
- Portfolio Management
- Customer Relationship Mgmt.
- Value / Outcome Reporting
- Value Gap Assessment

Semantic Governance (Service Management)

SEMANTIC GOVERNANCE

Service Delivery Model

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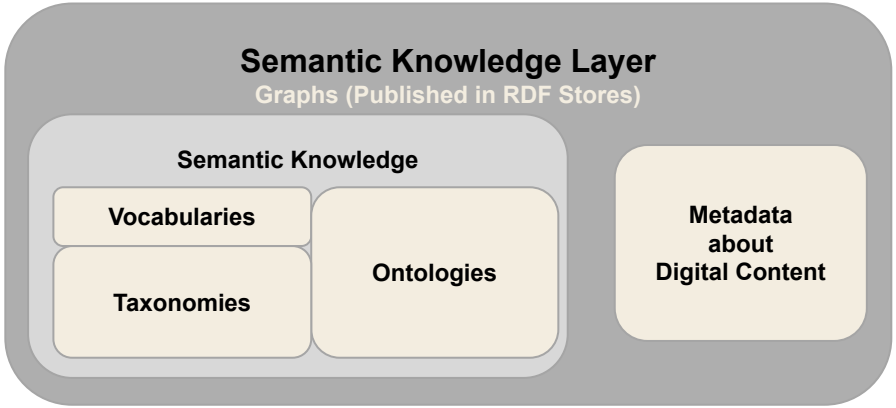
Service Operating Model

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What is Semantic Knowledge Layer (SKL) ?

A Semantic Knowledge Layer provides a business oriented cartography for navigating the expanding universe of digital assets and knowledge to:

- Make data easily findable and accessible for decision-makers by providing relevant insights and context at the right time.
- Enhance learning, curate context, and connect structured and unstructured data to help AI understand and interpret data and information to deliver meaningful insights.
- Support continuous learning for AI so it can adapt as new data, context, and information become available and improves insight delivery by enhancing transparency and explainability.

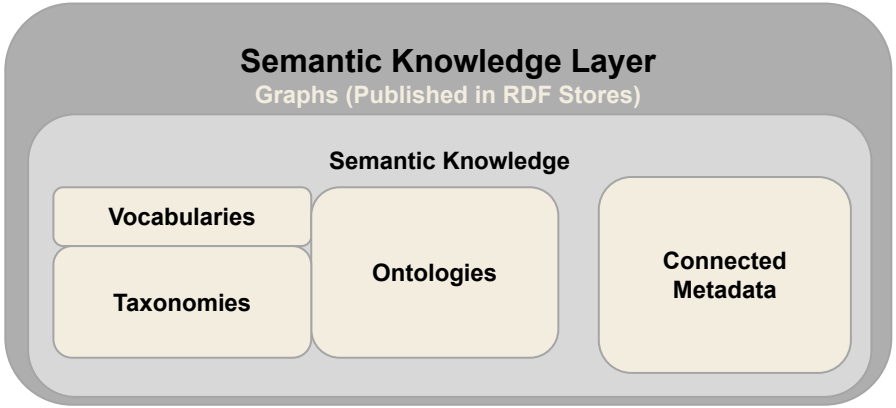


- Vocabularies - SKOS compliant controlled vocabulary that is reusable by taxonomies & ontologies
- Taxonomies - W3C standard structure with complex attribute management
- Ontologies - Specific POV approaches to data (business, Core) connected using flexible but standardized relationship management

What is Semantic Knowledge Layer (SKL) ?

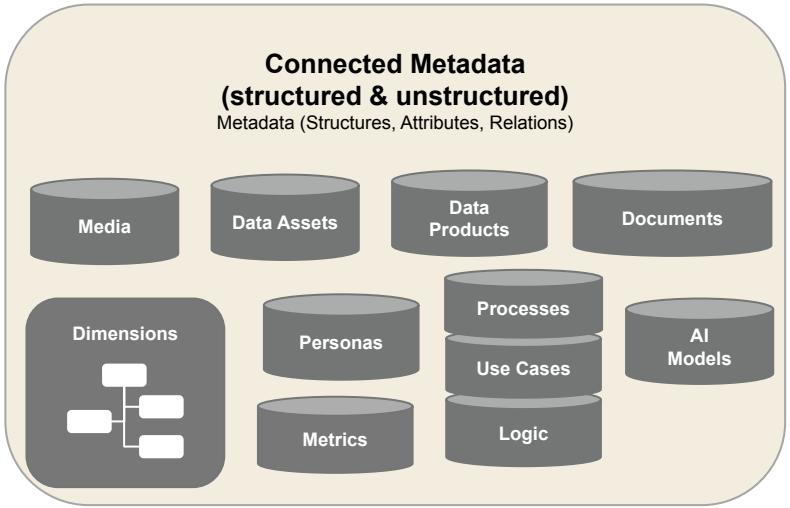
A Semantic Knowledge Layer provides a business oriented cartography for navigating the expanding universe of digital assets and knowledge to:

- Easily Search & Discover digital assets for decision-makers and automated agents
- Improve precision and accuracy for AI learning
- Provide relevancy using Personalization
- Support causal insights (WHY things happened, WHO is impacted, WHAT options are available)
- Improve transparency, explainability, and literacy



Components - Connected Metadata

Connected Metadata defines the structure, relationships, attributes, and definitions where possible about digital content and the manner in which it is stored. This is used for supporting technical services applied throughout the lifecycle of digital assets.



COLUMN METADATA

Name	Data Type	Description
cust_id_no	NUMERIC	The Unique Identifier For The Customer Id I

Attribute Name	Attribute Value
Data Element	Mapping of account number to customer
Security Class	Highly Confidential
Security Class Category	PII

MetaData Example

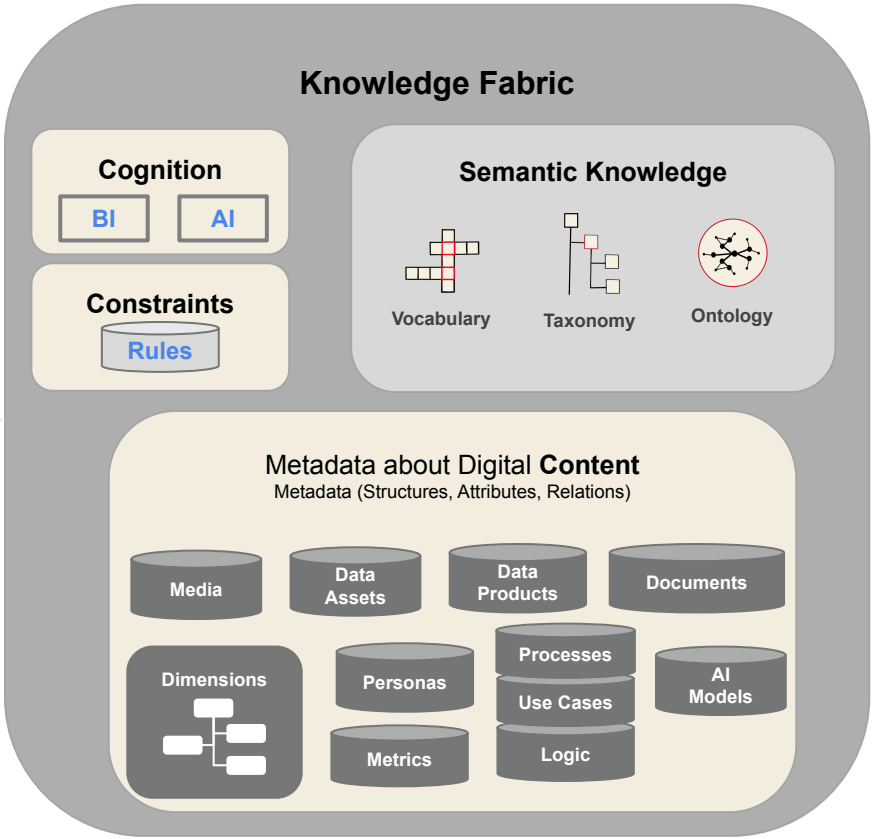
What is a Knowledge Fabric?

- Content (Structured: Relational Data)
- Content (Unstructured: Documents, Non-Relational)
- Context (taxonomies & ontologies: RDF, Graphs)*
- Cognition (Analytics = BI, Algorithms = AI)
- Constraints (Rules & Conditional Logic)

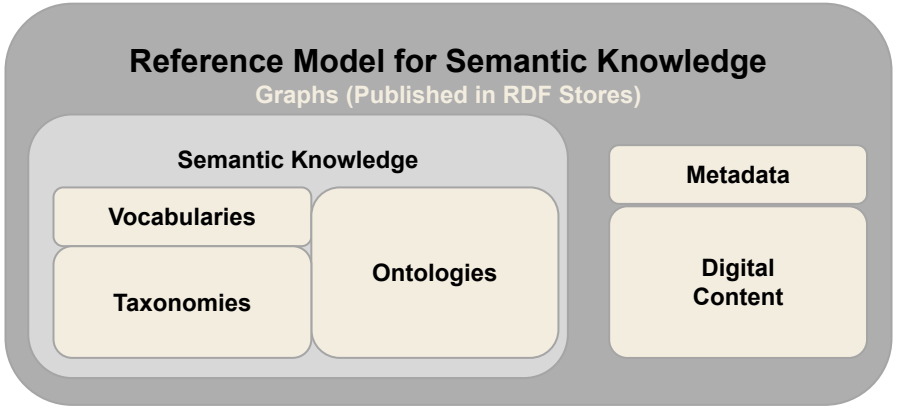




"Intelligent Assistant"



Conceptual View of Semantic Layer: Mapping Meaning to Digital Assets



Future

Building a Learning System

- Introduce lifecycle standards and design patterns for SI, AI & DI.
- Implement both virtual & physical knowledge graphs for better scale and efficiency.
- Develop an evolving learning capability by capturing clear facts (both relational and interconnected) & capturing probable knowledge (using vector-based methods and algorithmic improvements)

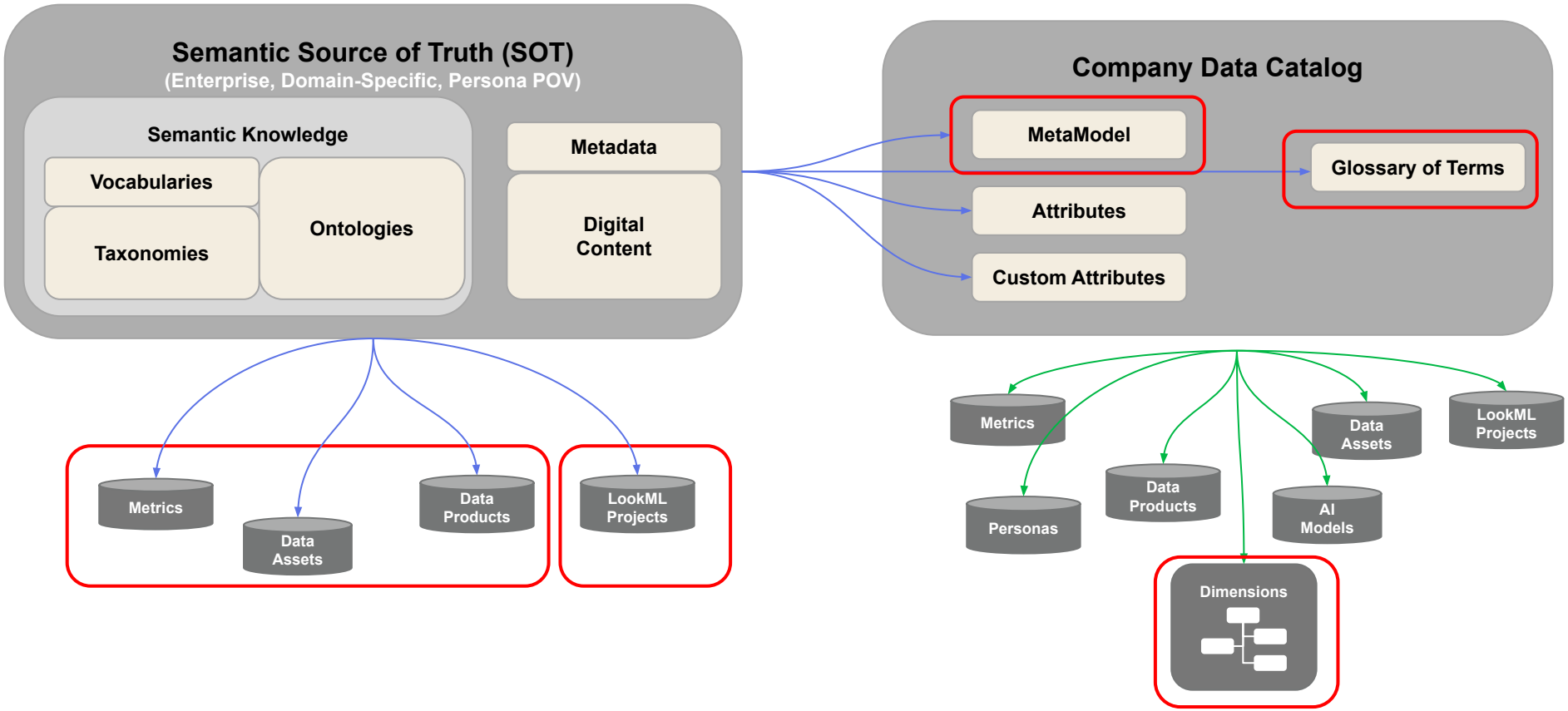
2025

The 3-Tier Approach to Smarter Content & Knowledge

Using our current infrastructure to make content more relevant to users based on their context and persona.

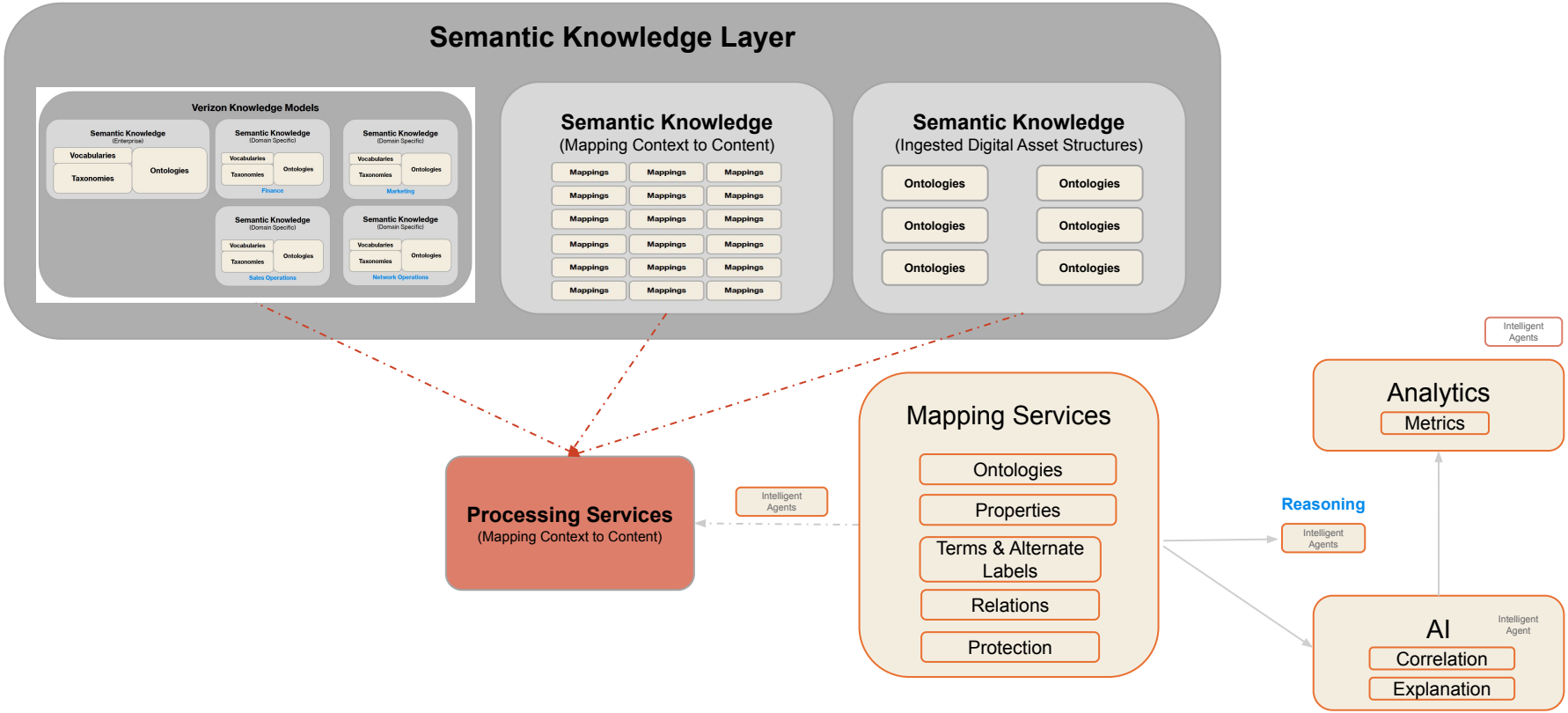
- Smarter organization by breaking down complex topics into manageable component knowledge, using processes as the core organizing principle. This involves building connections within and between different knowledge areas.
- Unified intelligence by establishing clear rules for semantic intelligence (SI) and align it with our artificial intelligence (AI), decision intelligence (DI) and business intelligence (BI) efforts leading to a more evolved model of our interconnected knowledge system as a design standard.
- Enhance search capabilities by finding data & objects, discovering relevant insights, finding direct answers to questions, offering suggestions during guided searches & developing an intelligent assistant to help users search better.

Mapping of Semantic Objects to Catalog Structures



Semantic Knowledge Layer: Semantic Services (Processing)

SEMANTIC GOVERNANCE



Future State

Semantic Knowledge Layer

SEMANTIC GOVERNANCE

What is it?
(Brief Description)

Structured representation of our knowledge, mapped across the Data & Analytics Ecosystem and integrated into a single, coherent whole.

It encapsulates the relationships, hierarchies, and semantic understanding of data.

This is crucial for effective diagnostics, accelerated decision-making, unified understanding vertically & horizontally, etc.

What is the value?
(Functional Features)

- The knowledge layer **provides the context in which metrics or KPIs operate.**
- By understanding the relationships between different data points, a knowledge layer can **help in determining the cause-and-effect relationships between metrics**.
- The knowledge layer can **suggest new KPIs or metrics based on emerging data patterns or business needs**.
- By understanding user roles and preferences, the knowledge layer can **customize metrics and reports based on the user's needs**.
- **Stakeholders can ask complex, natural-language questions** like "Why did our FWA sales drop last quarter?"
- When a change is detected (like a drop in a particular metric), the knowledge layer can **provide an impact analysis, highlighting other related metrics or business areas that might be affected**.
- The knowledge layer **allows users to dive deep into specific metrics for a granular view or aggregate data to get a higher-level perspective, catering to both micro and macro analytic needs**.
- If a metric deviates significantly from expected values, the system **can automatically search for potential reasons within the data, offering possible explanations (causal factors with weighting) for the anomaly**.

Confide

Future State

Positioning Semantic Knowledge

In Place | Extend | Build

SEMANTIC GOVERNANCE

Consumption Services

Adaptive UX + Unified Intelligence

- Dynamic Assembly of UI components
- UI Component Library
- Device Agnostic Facade (Screens & Forms)
- Service Library
- REST API Layer
- Synchronous Comm. Layer
- Analytic Framing (Contextual Relevancy)
- Visualization
- Std + Custom Dimensions

Synchronous | Asynchronous

Integration Services

- Domain Specific Services
- Shared Services
- Core Services
- kafka
- Process Models
- Service Library
- Search & Discovery
- Shared Services
- Provisioning & Protection
- Quality Profiling Services

Governance Services

- Metrics Library
- Dimensions
- SOT Catalog
- AI Registry (Engram)
- **Governance (Controls)**: Privacy | Protection | Provision | Sharing
- **Inventory Management**: Onboarding | Enriching | Classification | Publishing
- **Order Management**: Search & Discover | Test Data | Subscribe Prod Data
- **Operations (Data, Analytics, Systems, Platform)**: Monitoring & Measurement | Triage & Issue Management | Monitoring & Auditing of Controls

Data Aggregation & Curation Services

- Domain Specific Pre-Aggregates
- Domain Specific Shared Data Products
- Enterprise Shared Data Products
- Enterprise Pre-Aggregates

Data Management Services

- App DBs
- GCP
- Operational Data Stores

Prompts & NLP

Intuitive Search & Inquiry Services

Agents & Agentic AI

Correlation, Summary, Decision, Profiling, & Projection Services

Analytic Engines (Vendor Tools)

Semantic Services

Harmonized Semantics & Unified Relevancy

- Upper Ontology
- Taxonomies
- Personalization
- Causal Patterns
- Domain Ontologies
- Domain Ontologies
- Domain Ontologies
- Domain Ontologies

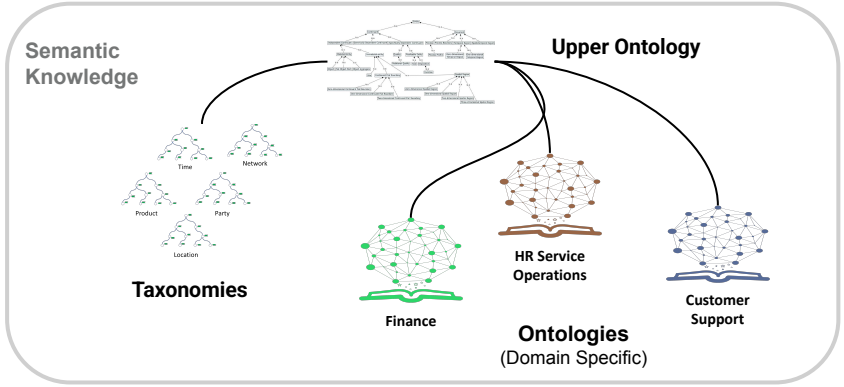
Semantic Metrics Layer

Unified Schema Model for Analytics & AI

Semantic Data Models
(Harmonized Structures)

How

Semantic Knowledge Layer (SKL) 3-Tier Architecture



Business Knowledge

R2R Mapping M:M

Metadata Structures

RDF Store
Persistent
(Triples for DDL)

Metadata

DDL 2 Data Mapping 1:1

Corporate Data
(Metadata + Instances)

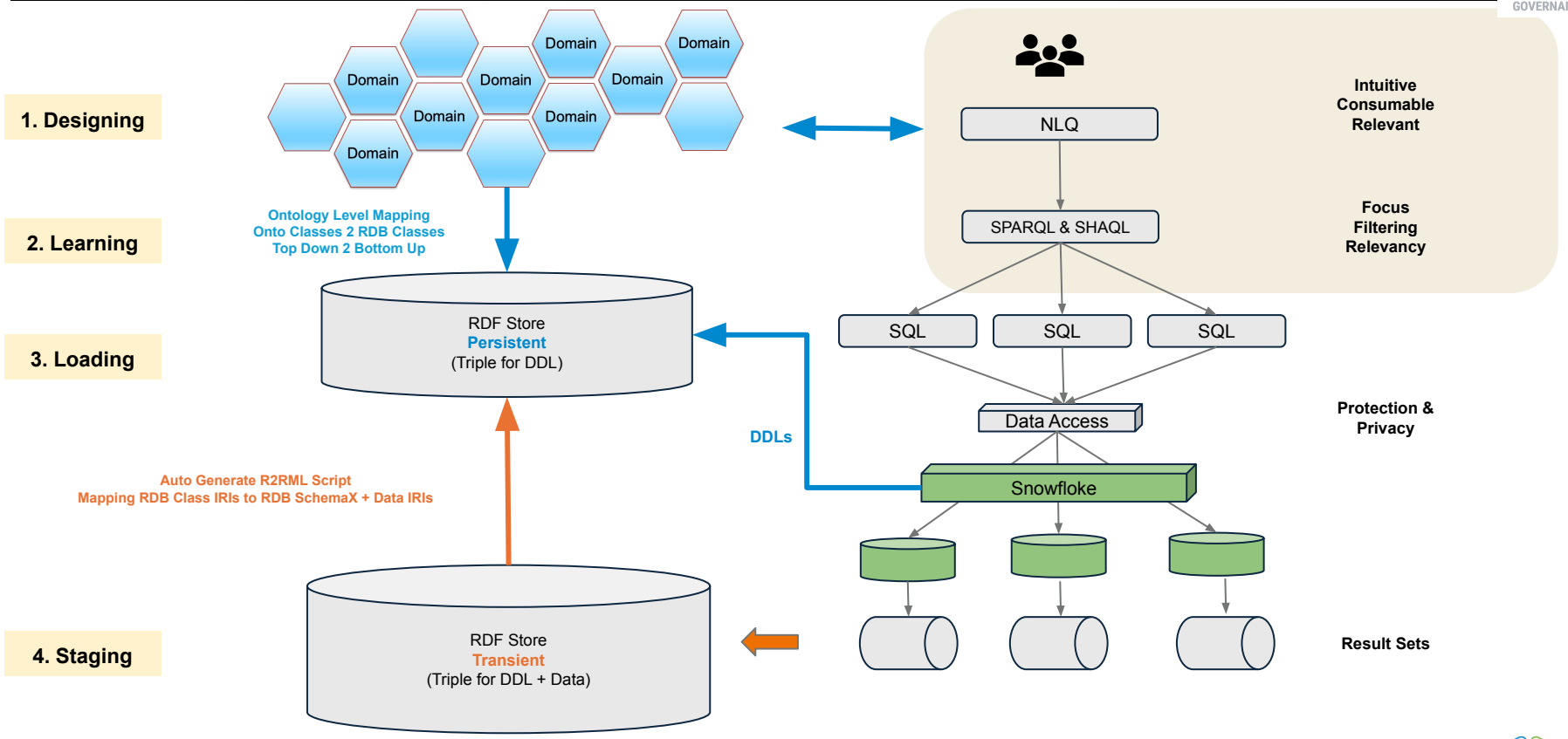
Content (Instances)

RDF Store
Transient
(Triple for DDL + Data)

Instances

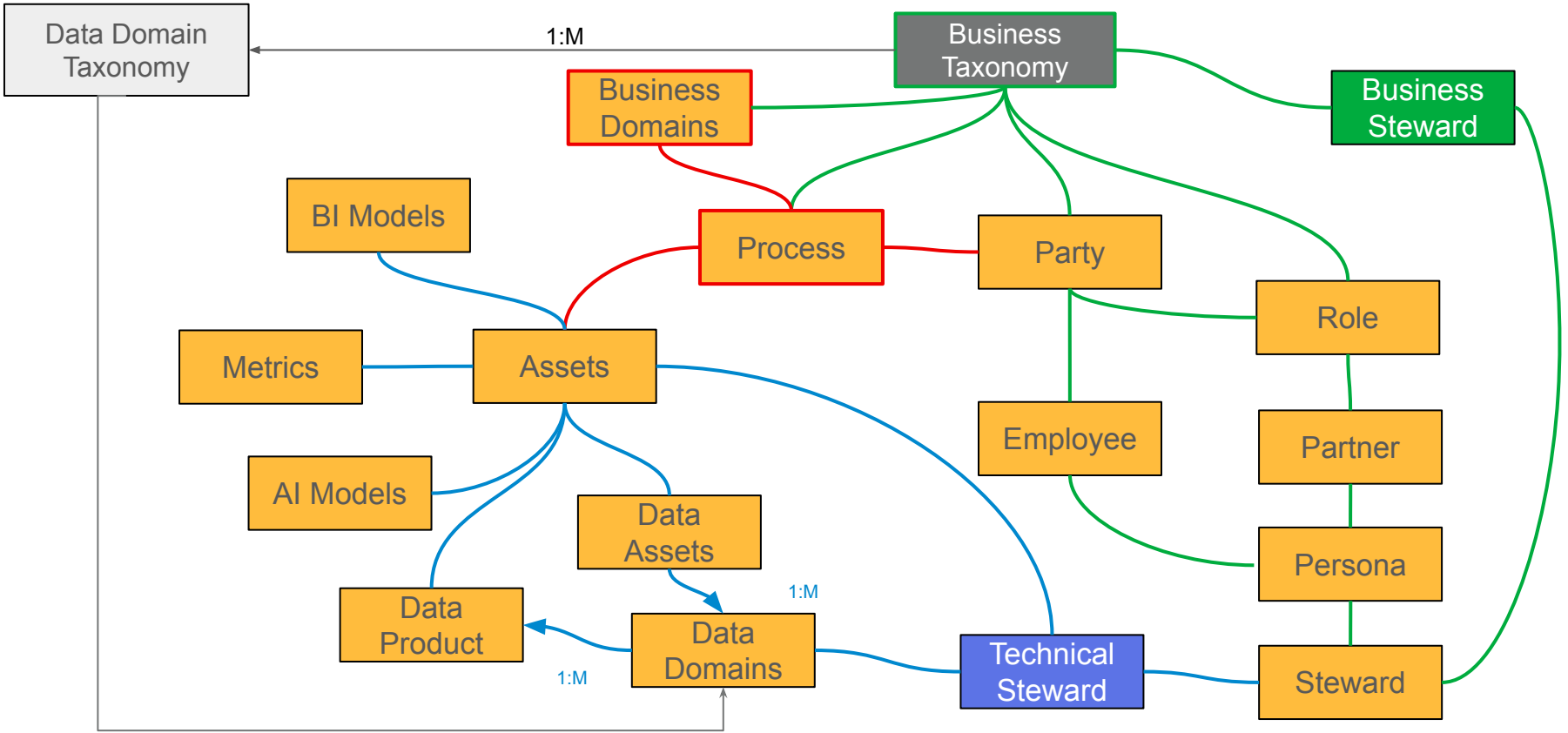
How

Semantic Pilot (Solution Architecture)



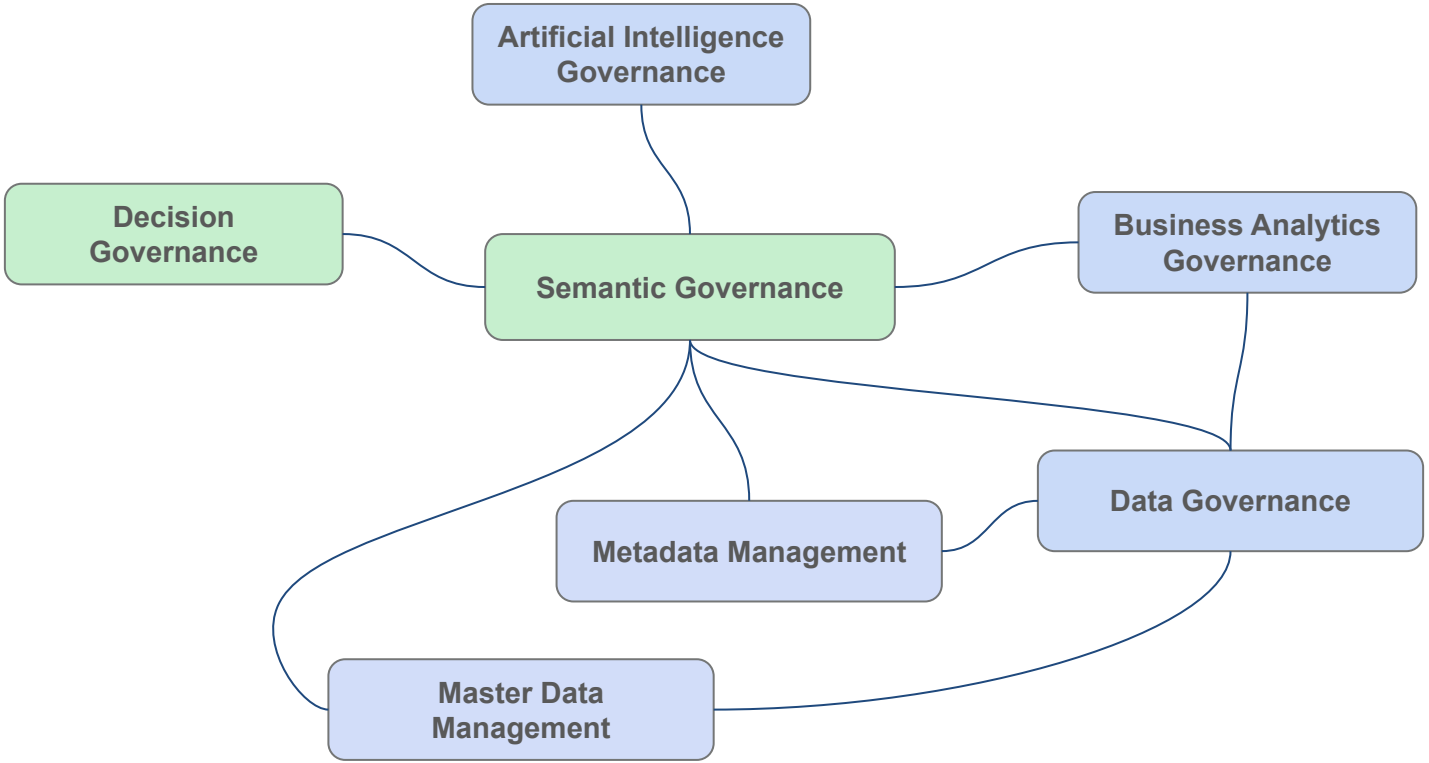
How

Semantic Governance (Business Taxonomy Framework)



How

Semantic Governance Alignment in Governance



How

Bottom Up Standard Naming Conventions

Content Types

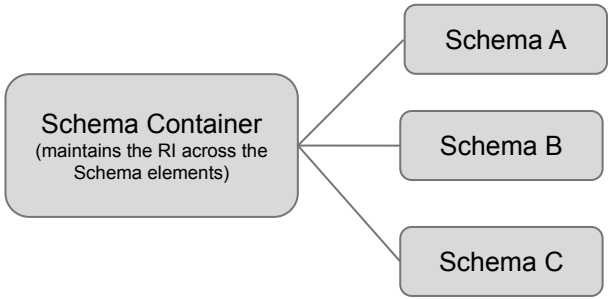
- Data Schemas
- Metric Schema(s)
- Semantic Layer for Analytic Models
- Document Schema(s)
- Hierarchy Schema(s)
- AI Model Schema(s)

Bottom Up Reference:

Schema | Container | Name

Bottom Up Reference:

Schema | Content | Name



How

Mapping Guideline

Content Types

- Data Schemas
- Metric Schema(s)
- Semantic Layer for Analytic Models
- Document Schema(s)
- Hierarchy Schema(s)
- AI Model Schema(s)

Bottom Up Reference:

Schema | Container | Name
Schema | Content | Name

Mapping Reference:

Mapping | Container | Name
Mapping | Content | Name

1. Identify which Top-Down Ontologies are included
2. Partition the mapping
   a. By Table (lowest unit of Content)
   b. By Container (View, DB, Project ID)
   c. Note: View has a (mostly) 1:1 Table

How

Mapping Guideline

